2021 IEEE Intelligent Transportation Systems Conference (ITSC)
Indianapolis, USA. September 19-21, 2021

2021 IEEE International Intelligent Transportation Systems Conference (ITSC) | 978-1-7281-9142-3/21/$31.00 ©2021 IEEE | DOI: 10.1109/ITSC48978.2021.9564754

RPFA-Net: a 4D RaDAR Pillar Feature Attention Network for 3D Object Detection

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Abstract*— 3D object detection is a crucial problem in environmental perception for autonomous driving. Currently, most works focused on LiDAR, camera, or their fusion, while very few algorithms involve a RaDAR sensor, especially 4D RaDAR providing 3D position and velocity information. 4D RaDAR can work well in bad weather and has a higher performance than traditional 3D RaDAR, but it also contains lots of noise information and suffers measurement ambiguities. Existing 3D object detection methods can't judge the heading of objects by focusing on local features in sparse point clouds. To better overcome this problem, we propose a new method named RPFA-Net only using a 4D RaDAR, which utilizes a self-attention mechanism instead of PointNet to extract point clouds' global features. These global features containing long-distance information can effectively improve the network's ability to regress the heading angle of objects and enhance detection accuracy. Our method's performance is enhanced by 8.13% of 3D mAP and 5.52% of BEV mAP compared with the baseline. Extensive experiments show that RPFA-Net surpasses state-of-the-art 3D detection methods on Astyx HiRes 2019 dataset. The code and pre-trained models are available at https://github.com/adept-thu/RPFA-Net.git.

I. INTRODUCTION

3D object detection is one of the most challenging problems in the perception system of autonomous driving. The weather adaptability, detection distance, and price of LiDAR restrict the application of 3D object detection. In contrast, RaDAR makes up for the defects of LiDAR, and it is a sensor with more commercial prospects.

3D RaDAR can detect the horizontal position and velocity of objects, usually expressed as point clouds. To adept the input format of the deep learning network, the data can be divided into grids or converted into bird's eye view images.

This work was supported by the National High Technology Research and Development Program of China under Grant(No. 2018YFE0204300), the Beijing Science and Technology Plan Project (Z191100007419008), the Guoqiang Research Institute Project (2019GQG1010), the National Natural Science Foundation of China under Grant (No. U1964203), the Funding for National Defense Basic Research Program (No.6142006190201) and the China Postdoctoral Fund(No.2021M691780).).

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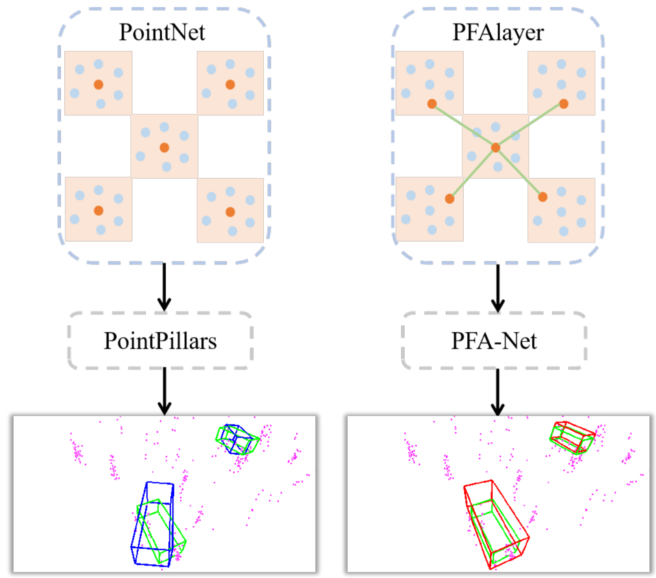

Fig. 1. Advantages of RPFAlayer. The squares in the figure represent pillars, the blue dots are point clouds, the orange dots are extracted features, and the green straight lines represent the connections among the pillar's global features. The detection results are displayed at the bottom. The pink points are 4D RaDAR points cloud, the green boxes are ground truth, RPFA-Net gives the red boxes, and PointPillars gives the blue boxes. PointNet extract the point cloud features that best describe this pillar, but there is no connection between pillars. RPFAlayer uses self-attention to capture the relationship between pillars and selects the elements that best represent the global relationship. This method improve the regression ability to head angles and enhance the accuracy of object detection.

However, the sparsity of the RaDAR point cloud is still a challenge. The appearance of PointNet [2] provides a possibility for processing sparse data. Andreas Danzer et al. [14] use Frustum PointNets [20] to classify, segment, and regress 3D RaDAR. The advantage of this method is that it does not need to manually design features or mesh the point cloud and can directly take the point cloud as input. However, 3D RaDAR data lack vertical information, and it can not use single-mode work in 3D object detection scene.

Many researchers fuse 3D RaDAR with other sensors and add 3D RaDAR point cloud as supplementary information to the detection network. For example, CenterFusion [4] returns the center point of the object in Centernet [5]. It utilizes a truncated way to fuse RaDAR information to supplement the image features and regresses the depth, direction, and velocity of objects. In this method, RaDAR is only an auxiliary sensor to provide the depth information and speed information, and the camera carries out the main work. Besides, RadarNet [21] is a fusion detection method of LiDAR and 3D RaDAR, which can detect dynamic objects based on early voxel fusion and late attention fusion. But its pre-fusion extracts voxel features from point clouds with different properties and then uses channels to splice them. This method ignores the difference of point cloud between 3D RaDAR and LiDAR. Therefore, both industry and science are actively introducing 4D RaDAR sensors. 3D RaDAR can only detect the horizontal position of objects, while 4D RaDAR sensors can detect the absolute 3D coordinates of objects. Consequently, it is imperative to propose suitable 3D object detection methods for 4D RaDAR.

However, the 4D RaDAR sensor appears late, and there are few data sets. As far as we know, the only data set including RaDAR data at present is nuScenes [3], but the 3D RaDAR used by this sensor only contains sparse 2D points, and there are only about 100 points per frame. Therefore, it is a challenging task to identify different objects by only using RaDAR data. If the 4D RaDAR point cloud with 3D coordinates and velocity information can be directly processed, then the method is similar to the technique of LiDAR from the above work. In the mature end-to-end 3D object detection methods based on LiDAR [15], [17], [18], [19], point clouds are usually transformed into voxel features, or features are extracted directly from point clouds. Although we can perform the same operation on 4D RaDAR point cloud, the local features obtained by the above methods do not perform well in sparse data.

With the development of autonomous driving and sensor technology, it emerges new 4D RaDAR sensors and a 4D RaDAR dataset, Astyx HiRes2019 [6], which contains 3D position and speed information of each object. The 4D point cloud in this dataset can measure the position and contour of objects just like LiDAR point cloud, and each point contains velocity information. Since most works focus on LiDAR-related work, works using this data set are very rare. The principle of RaDAR and LiDAR sensors is different, and their points cloud have wide variations. The 4D RaDAR point cloud is often sparser and has many noises. Besides, due to the weak ability to describe the object's shape, it can not judge the object's orientation through the point cloud of a single object. The existing 3D object detection methods developed from the LiDAR dataset unable to adapt to the 4D RaDAR data set.

This paper proposes a 4D RaDAR-based 3D object detection network RPFA-Net using the self-attention mechanism to solve the above problems. RPFA-Net utilizes the self-attention mechanism to extract point clouds' global features and it judges objects' orientation through the global characteristics. Because the orientation angle is more accurate, the coincidence between the detection results and the ground truth will be higher, effectively improving detection accuracy. We conduct model training and experimental verification on the 4D RaDAR data set Astyx HiRes2019 [6]. The experimental results show that the detection accuracy of RPFA-Net is greatly improved compared with our baseline.

The structure of this work is as follows. Section II introduces the related work of RaDAR on object classification, semantic segmentation, and object detection. Section III shows our proposed 3D object detection method. Section IV presents the data set and training environment. Section V gives the experimental evaluation results and is followed by conclusions in section VI.

II. Related Work

A. Object Classification based on RaDAR

In the aspect of RaDAR data classification, the work [7] adopts CNN (Convolutional Neural Networks) to obtain the semantic information of RaDAR data. It proves that the network can get semantic information from the RaDAR grid by classifying its contained objects. The method can directly affect the semantic information in the divided grid to improve the network's reasoning efficiency. In the paper [8], the deep neural network is used to learn the RaDAR spectral data, and the classification effect is better than that of traditional KNN (K-Nearest Neighbor) and SVM (Support Vector Machine). The above methods only fit static object detection, while the dynamic objects need to be detected for autonomous driving. This paper [9] is based on Random Forest and Long Short-Term Memory to classify dynamic RaDAR objects in the moving object classification aspect. The paper [10] has a deep neural network composed of recursive convolutional units to classify dynamic RaDAR objects.

B. Object Segmentation based on RaDAR

RaDAR data and image data are different in terms of data sparsity. The non-zero value in RaDAR data only accounts for a small part. However, the efficiency of the image segmentation algorithm is very low, and the effect is poor. PointNet++ [11] is utilized to extract the semantic features of RaDAR data, and the vector generated by clustering reflection is regarded as the input to segment the RaDAR data. The work [12] proposes a method to deal with sparsity and noise using RaDAR data for road segmentation. This method uses multi-frame aggregation to learn an ISM(Interpretative Structural Modeling) function from the data. Also, the above method uses LiDAR data for supervised learning to solve the problems related to the distance and coverage difference between LiDAR and RaDAR.

C. Object Detection based on RaDAR

At present, the work of object detection in a single RaDAR sensor is relatively few. We list some typical works. RRPN [13] is a deep learning method for object detection of RaDAR data by utilizing the region proposal network. It projects RaDAR data on the image and then generates anchor bounding boxes for each RaDAR point. This method is based on mature techniques in the picture for reference, transforming RaDAR into a similar image format to use the image object detection method. To avoid the loss of RaDAR information in projection, PointNet acts the feature extractor and estimates the 2D bounding box of the object [14].

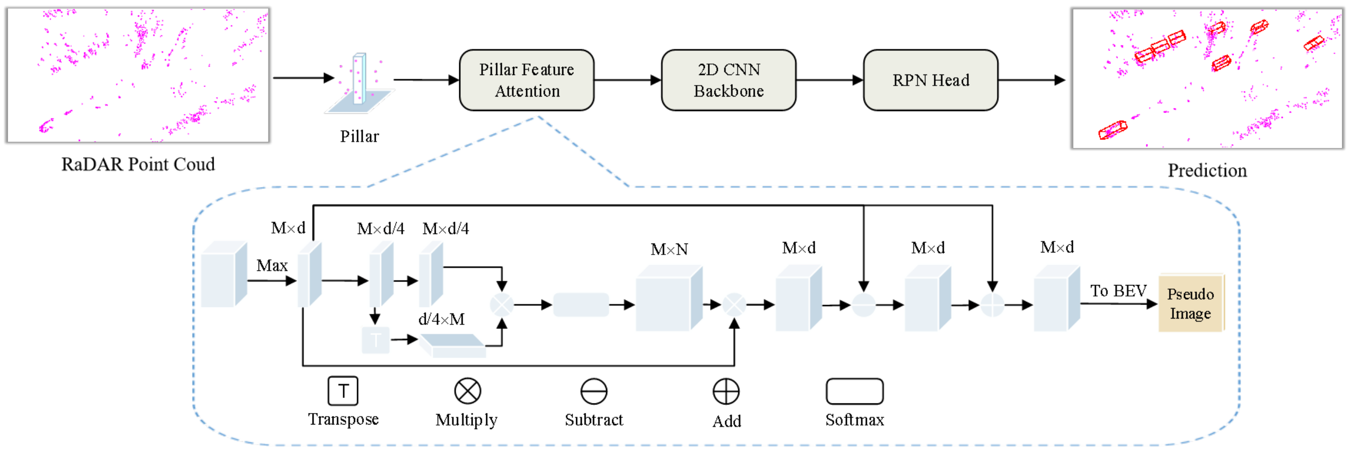

Fig. 2. Network architectures for RPFA-Net. The blue cubes represent the tensors of RaDAR Pillar features, and the numbers above represent the number and dimensions of features in the tensor.

We can observe that most methods about 3D RaDAR, and there is no 3D object detection based on end-to-end 3D RaDAR. Because of the lack of information in 3D RaDAR data itself, it can't work independently in 3D object detection tasks. It can only carry out 2D object detection or serve as supplementary information for other sensors. Therefore, we propose an end-to-end network RPFA-Net for 3D object detection directly using a single 4D RaDAR sensor. It uses self-attention to extract the global features of point clouds instead of the traditional PointNet, and finally, the RPN head returns to the 3D bounding box of the object.

III. Pillar Feature Attention Network

The structure of RPFA-Net using 4D RaDAR for 3D object detection is shown in Fig. 2, which is composed of Pillar Features Attention module, 2D CNN backbone network, and RPN detection head. Pillar Features Attention extracts the global features of the point cloud through the attention mechanism and then generates detection results through a 2D CNN backbone network and RPN detection head. In this section, the overall design of RPFA-Net and the feature extraction layer RPFAlayer based on the self-attention mechanism will be introduced in detail.

A. RPFA-Net

To consider both accuracy and efficiency, we select PointPillars [15] as our baseline. PointPillars is an end-to-end object detection network using LiDAR point cloud as the input. It practices the grid-enabled method to divide the point cloud into pillars and then extract features from each pillar's points. Finally, it encodes these features into a multichannel pseudo image. The advantage of the processing point cloud as a pseudo image is that it can use efficient 2D CNN to process the pseudo image, maximizing network reasoning speed while ensuring accuracy. However, the feature extraction of point clouds by PointPillars focuses on the local features and can not judge the orientation of the object by the global attributes. So we utilize the self-attention mechanism to extract point clouds' part, that the pseudo image gets the global features. After passing through the leading 2D CNN network of PointPillars to output the pseudo image features, the detection results are obtained by RPN the head finally.

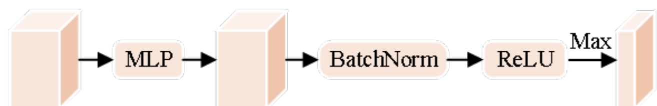

Fig. 3. A simplified version of PointNet structure used in PointPillars. The input features are simply coded low-dimensional features in Pillar, and the output features are high-dimensional features.

B. RPFAlayer

The structure of extracting point cloud features from PointPillars is shown in Fig. 3, which comprises an MLP, a Batchnorm, a Relu function, and a maximum operation. It can be regarded as a simplified PointNet structure. The point cloud features extracted by it belong to local features, which is not conducive to the regression of the measured object's direction angle. Since it is difficult to judge the direction of an object in the 4D RaDAR point cloud, we should capture the context information as much as possible and estimate the object's direction by establishing a long-distance feature dependency. Self-attention mechanism[16] is a method to capture context information. It maps the original feature graph into three vector branches, namely query, key, and value. Firstly, the correlation weight matrix coefficients of question and key are calculated; secondly, the weight matrix is normalized by smooth operation; finally, the weight coefficients are superimposed on the value to build the modeling consisting of global context information. To apply the self-attention mechanism to feature extraction of the point cloud, we design RPFAlayer to extract the point cloud's global context features. To improve the network's efficiency and reduce the training parameters, we first perform the maximum operation on the input feature, which is equivalent to finding the feature that best represents the pillar. Then, the feature is mapped into three components:

Q, K, and V, and only attention operation is performed on them. Finally, the feature with global context information is residually connected with the origin residual to extract the point cloud feature. The specific implementation is shown in the following equations:

$$\begin{cases} F_{64} = Linear(f_{max}(F)) \\ F_{16} = Conv(Linear(f_{max}(F))) \\ F_w = Softmax((F_{16})^T F_{16}) \\ F_t = ReLU(BN(LN(F_w F_{64} - F_{64}))) \\ F_{out} = F_t + F_{64}, \end{cases} \quad (1)$$

where F is the characteristic vector of input RPFAlayer and $f_{max}()$ is the maximum operation in the pillar, and linear and $Conv()$ is the linear and convolution layer. F_{64} is the characteristic matrix before dimension reduction, and F_{16} is the characteristic vector after dimension reduction. Dimension reduction is mainly for reducing parameters and improving efficiency. F_w is the attention weight matrix. $Softmax()$, $BN()$ and $Relu()$ denote the normalized function the batch normalization layer and the activation function.

C. Data Preprocessing

There is a time difference between the electromagnetic wave emitted by the 4D RaDAR sensor and the electromagnetic wave reflected by the measured object, which calculates the point cloud's position according to it. The 4D RaDAR point cloud should describe the object's shape and position precisely as the LiDAR point cloud. However, there is a divergence in the vertical plane of the data, that the height measurement of the data is not accurate. Because of the angular resolution problem of the sensor, an object within an angle may only be expressed as a point. If there are other points in the object, they will appear in other places due to the angular resolution problem. We propose a data preprocessing method of 4D RaDAR point cloud data based on statistical law to solve this problem. First, the divergence angles of each point in each frame are counted to see if they conform to the normal distribution. If so, the average value is selected to represent the divergence angles of all points in this frame. Otherwise, the model is used to represent them. The angle calculated above is employed to adjust the vertical plane coordinates of the point cloud, namely X and Z coordinates. The specific method is as follows:

$$\tilde{\theta} = \sum_{i=1}^{n} \theta_i / n, \quad (2)$$

$$\begin{cases} x_t = cos(\frac{2\theta_t}{\theta_d} arctan\frac{z}{x})\sqrt{x^2+z^2} \\ z_t = sin(\frac{2\theta_t}{\theta_d} arctan\frac{z}{x})\sqrt{x^2+z^2}, \end{cases} \quad (3)$$

where θ_t, $\tilde{\theta}$ represent divergence angle, average value and variance. θ_t, θ_d denote object angle to be compressed and source angle before compression. x and z are the source coordinate of the 4D RaDAR point cloud, while x_t and z_t are the adjusted coordinates.

IV. Experiments

This section describes the evaluation results of RPFA-Net on the Astyx HiRes2019 data set. For that, we compare the detection accuracy of RPFA-Net with some existing networks and show the experimental ablation results for data preprocessing and RPFAlayer. Besides, we introduce the dataset, experimental platform, and parameter configuration.

A. Evaluation Results

B. Dataset

The dataset used in this work is Astyx HiRes2019[6], which is jointly collected by a 4D RaDAR sensor Astyx 6455 HiRes, a LiDAR Velodyne vlp-16, and a Point Grey Blackfly camera. It consists of 546 synchronous frames, including more than 3000 3D object annotations. There are seven categories in the labeled data: car, bus, human, bicycle, motorcycle, truck, and trailer. The label of car accounts for the majority. The label document contains the 3D position, 3D rotation, 3D size, category, occlusion, and uncertainty of the annotated object. The number of 4D RaDAR and LiDAR point clouds in the data set is 1000 $\sim$ 10000 and 10000 $\sim$ 25000 respectively, while the number of pixels in the image is 2048 $\times$ 618.

Because the number of samples in the data set is too small, we choose the self-help method to divide the 546 frame data set into 410 frames as the training set and 136 frames as the validation set using a ratio of 3:1. We are keeping the consistency of data distribution as much as possible and avoiding the final result due to additional bias in the data division process. There are 2204 cars, 36 pedestrians, and 8 cyclists in the final training set. Because the other two labels are too small, our final evaluation result are only for the car category.

C. Training and Parameter

The training platform is a single Nvidia RTX3090 GPU, and the training time is about two and a half hours. When dividing voxels, the length, width and height of each voxel are [0.16, 0.16, 4], the X-range , Y-range , Z-range of the point cloud is [0, 99.84], [- 39.68, 39.68] and [- 3, 1]. Finally, the point cloud is divided into 624 $\times$ 496 voxels.

We compare the results of RPFA-Net with PVRCNN [17], PointRCNN [18], SECOND [19] and PointPillars [15], which demonstrate superior performance on the KITTI data set. As shown in Table I, we can see that PointPillars have the best performance on the data of single-mode RaDARs. RPFA-Net is superior to all the above methods, which is 8.13% higher than PointPillars in 3D mAP and 5.52% of BEV mAP. We compare PointPillars and RPFA-Net separately in Table II. From the comparison of AOS in the table, RPFAlayer can greatly improve the direction angle regression accuracy of the bounding box, the accuracy of the direction angle, 3D mAP, and BEV mAP. To verify the effectiveness of data preprocessing and RPFAlayer, we design an ablation experiment that data preprocessing and RPFAlayer can effectively improve the detection accuracy from Table III.

TABLE I

COMPARATIVE EXPERIMENT OF THE FOUR NETWORKS.

Modality	Methods	3D mAP(%)			BEV mAP(%)		
		Easy	Moderate	Hard	Easy	Moderate	Hard
RaDAR	PointRCNN[18]	12.23	9.10	9.10	14.95	13.82	13.89
	SECOND[19]	24.11	18.50	17.77	41.25	30.58	29.33
	PVRCNN[17]	28.21	22.29	20.40	46.62	35.10	33.67
	PointPillars[15]	30.14	24.06	21.91	45.66	36.71	35.30
	RPFA-Net(our)	**38.85**	**32.19**	**30.57**	**50.42**	**42.23**	**40.96**

TABLE II

COMPARATIVE EXPERIMENT OF POINTPILLARS AND RPFA-NET FOR AVERAGE ORIENTATION SIMILARITY.

Modality	Items	Pointpillars			RPFA-Net(our)		
		Easy	Moderate	Hard	Easy	Moderate	Hard
RaDAR	BBOX	19.05	17.12	15.96	54.04	45.76	43.77
	BEV	45.66	36.71	35.30	50.42	42.23	40.96
	3D	30.14	24.06	21.91	38.85	32.19	30.57
	AOS	18.86	16.96	15.85	**53.26**	**44.98**	**42.91**

TABLE III

4D RADAR DATA PREPROCESSING AND RPFALAYER ABLATION EXPERIMENTS.

Methods	3D mAP(%)			BEV mAP(%)		
	Easy	Moderate	Hard	Easy	Moderate	Hard
RaDAR	30.14	24.06	21.91	45.66	36.71	35.30
RaDAR+data process	32.95	26.14	25.11	50.35	40.05	37.77
RPFAlayer	37.14	30.15	28.3	49.29	30.14	39.48
RPFAlayer+data process	**38.85**	**32.19**	**30.56**	**50.42**	**42.23**	**40.96**

Fig. 4. Qualitative results on the Astyx HiRes2019 data set. The RaDAR point cloud is pink. The bounding boxes of ground truth are green, the bounding boxes of PointPillars are sky blue, and the bounding boxes of the RPFA-Net are blue. The indigo dashed circles are false detections, and the orange dashed circles are missed detections.

Fig. 4 shows the results detected by RPFA-Net and PointPillars, and we circle the false detection caused by PointPillars with a green dotted line. Because of the 4D RaDAR point cloud's sparsity, it is impossible to judge the measured object's orientation by local point cloud features. An accurate object-orientation angle can make the detected bounding box coincide with the ground truth to the greatest extent to get more accurate detection results. Since most vehicles are driving on the highway, most vehicles' direction angles are not much different. When we use RPFAlayer to capture point clouds' global information, multi-tiple adjacent features can better judge objects' orientation angle. The local features extracted by PointNet in Fig. 1 only locate the center position of the measured object after learning by PointPillars network, but the heading angle has excellent deviation. After RPFAlayer extracts global features, RPFA-Net can correct these deviations to a great extent. In the third group of pictures in Fig. 4, we can also see that RPFA-Net regression's heading angle is more accurate. We can also see from other visualization results in Fig. 4 that RPFA-Net can obtain more accurate heading angles, reject some false detections, and detect missing objects.

In addition to visualization of object detection results, our third table compares RPFA-Net and four existing point cloud detection networks, the comparison between RPFA-Net and PointPillars, and the ablation experiments RPFAlayer and data preprocessing. Table III shows that in the existing point cloud detection network, PointPillars has the best detection effect for 4D RaDAR. Compared with PointPillars, RPFA-Net has 8.13% improvement of 3D mAP and 5.52% improvement of BEV mAP. The AOS in Table II represents the Average Orientation Similarity, and the AOS of RPFA-Net overperform PointPillars by a large margin, with a maximum range of 28.02%. Our ablation experiments in Table III show the effectiveness of data preprocessing and the improvement of RPFAlayer on detection accuracy. To sum up, RPFA-Net achieves the state of the art accuracy in 3D object detection only using a 4D RaDAR sensor.

V. Conclusions

In this paper, we propose a 4D-RaDAR 3D object detection network RPFA-Net based on the self-attention mechanism. The current methods can not extract global features well and estimate the heading of the bounding box reasonably. We use the self-attention mechanism instead of the simplified PointNet to extract the point cloud features in the pillar and propose RPFAlayer, which can effectively capture the information in a long distance. Experiments show that RPFAlayer can significantly improve the accuracy of network regression heading angle and improve the accuracy of 3D object detection.

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